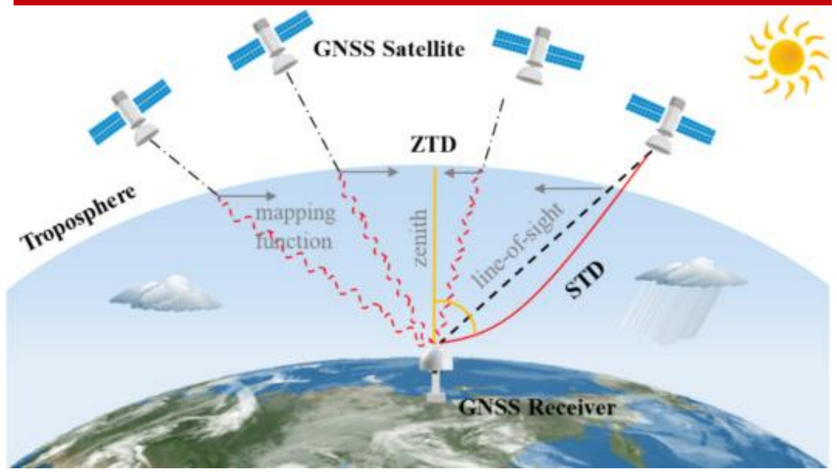
ECE SPARK 2026

Atmospheric Sensing Using Global Navigation Satellite System

Wenmiao Jia, Parmesh Ramanathan, Bhuvana Krishnaswamy

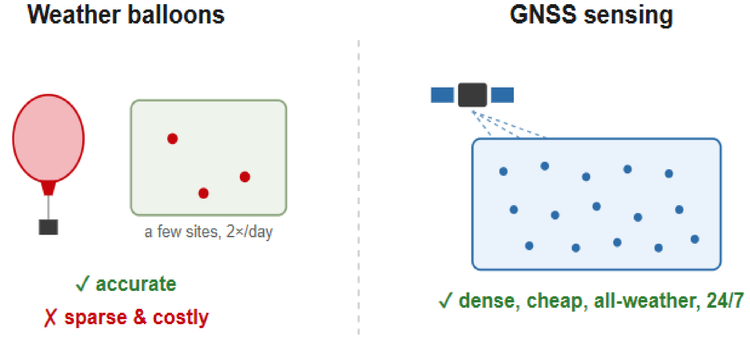
Background



- ZTD: zenith tropospheric delay
- STD: slant tropospheric delay

Fig. adapted from Lu et al., *Earth-Science Reviews*, 2026.

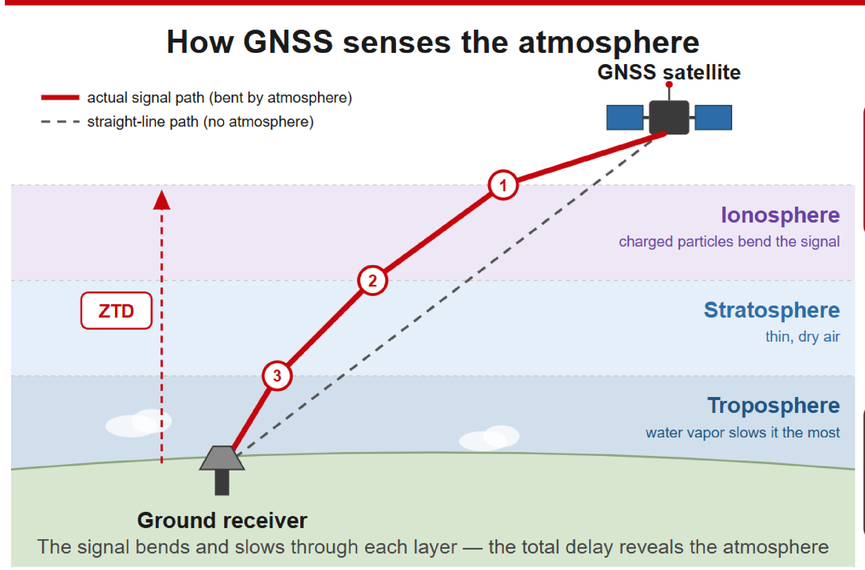
- Water vapor drives weather, and GNSS satellite gives dense, cheap, weather sensing
- Measure atmospheric water vapor from Global Navigation Satellite System signals (GNSS)
- Weather balloons: accurate, but sparse and costly
- GNSS sensing: well established L-band(1.5GHz)

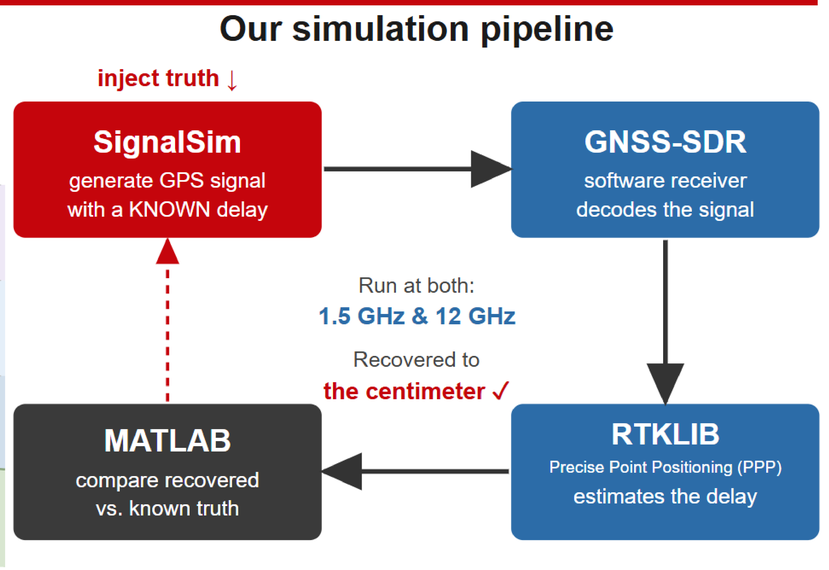


Objective

- Recover the tropospheric delay to centimeter accuracy

Approach / Methodology





Did we recover the delay we put in? → yes

- Injected a known delay, so we can measure the exact error every run
- Swept many delay levels – recovery is linear and unbiased
- Held the ionosphere fixed so it couldn't skew the estimate
- Cross-checked against a real GNSS station(EarthScope/GAGE) – matched to ~0.1 m

Results and Conclusions

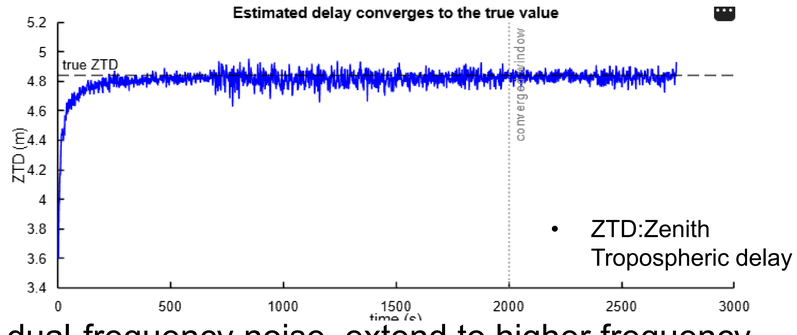
Delay recovered — two independent methods

True delay = 2.640 m

Method	Recovered	Error	Spread (±1σ)
Single-frequency (L1)	2.648 m	+8 mm	±52 mm
Dual-frequency (iono-free L1+L2)	2.661 m	+21 mm	±255 mm

Both land within ~2 cm of truth

single-frequency is the more precise (dual-frequency amplifies noise)



- ZTD:Zenith Tropospheric delay

- Pipeline built and validated, delay recovered. Reduce dual-frequency noise, extend to higher frequency, GNSS measures water vapor single frequency is enough
- Supports GNSS as a cheap, reliable atmospheric sensor going forward
- Vision: extend this to Ku-band (~12 GHz) signals from Low Earth Orbit (LEO) satellite constellations enables much denser higher-frequency atmospheric sensing